INCH POUND

MIL-S-38043A(USAF)
20 FEBRUARY 1995

SUPERSEDING
MIL-S-38043(USAF)
8 NOVEMBER 1962

MILITARY SPECIFICATION

STARTER, ENGINE, AIR TURBINE STU-16/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of pneumatic air turbine engine starter, designated STU-16/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
TT-E-489	Enamel, Alkyd, Gloss, (For Exterior And Interior Surfaces)
TT-P-1757	Primer Costing, Zinc Chromate, Low-Moisture-Sensitivity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2995

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Military

MIL-P-116	Preservation, Method Of
DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5400	Electronic Equipment, Aircraft, General Specifications For
MIL-T-5624	Jet Fuel, Grades JP-3, JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-L-7808	Lubrication Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-P-19834	Plate, Identification, Metal Foil, Adhesive Back
MIL-N-25027	Nut, Self-Locking, 250° F, 450° F, And 800° F, 125 KSI FTU, 60 KSI FTU, And 30 KSI FTU
MIL-L-27502	Lubricating Oil, Aircraft Turbine Engine, Ester Base
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-480	Configuration Control
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Specifications And Standards, Order Of Precedence For The Selection Of
MS20995	Wire, Lock
MS33540	Safety Wiring, General Practices For

Air Force-Navy Aeronautical

AND10262	Flange-Type X Accessory Mounting
AND10476	Washer-Limitations On Usage Of Lock
AND20002	Drive-Type XII Engine Accessory

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are

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those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)
ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the even event of a conflict between the test of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The starter shall consist of a turbine, reduction gearing, and engaging mechanism, a spline drive to mate with the engine drive, and control equipment.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3.1 Model specification. The contractor shall prepare a model specification in accordance with the requirements specified herein and submit it to the procuring activity for approval prior to production testing.

3.4 Materials. Materials shall be suitable for use with air or any combination of air and combustion products of fuel conforming to MIL-G-5572, grade 100/130, and MIL-J-5624.

3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service. Metals shall be inspected in accordance with FED-STD-151.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.1.2 Recovered materials. Recovered materials are used to the maximum extent possible without jeopardizing the end use of the item.

3.4.2 Protective treatment. When materials are used in the construction of the starter that are subjected to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent

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compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.3 Fungus-proof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be protected or treated with a fungicidal agent approved by the procuring activity.

3.5 Design. The starter and control valve shall be designed to operate on compressed air from a source such as a gas turbine power unit or as otherwise specified. The air consumed shall be the minimum amount commensurate with the performance requirements. The valve pressure drop when in the wide open position shall be held to a minimum.

3.5.1 Service life. The starter drive components shall be adequate or a starter delivery of 720 pound-feet maximum design torque and shall incorporate a drive shear section that will fail at a static torque not exceeding 1,050 ±50.

3.5.3 Starting. The starting requirements for the gas turbine engine shall be as shown on the torque versus rpm curve FIGURE 1. The starter output shall not exceed the maximum torque shown on FIGURE 1. The minimum torque shall be that obtained from the match between the auxiliary power unit conditions specified in 3.7.3 and the starter effective nozzle area.

3.5.4 Effective starting inertia. The effective mass moment of inertia at the starter drive, of the engine (not including accessories) masses to be rotated by the starter, is 19.2 slug feet squared.

3.5.5 Valve override closing device. Design shall be such that the load control and shutoff valve will not open unless the valve solenoid is energized. A mechanical malfunction of the valve induced by vibration or other conditions after a main engine start shall not allow the valve to open and permit air to be transmitted to the pneumatic starter.

3.5.6 Fittings. Quick disconnect fittings shall be provided for both entrance and exit connections.

3.5.6.1 Engine attachment. Attachment of the starter to the gas turbine engine shall not require any modification of the engine. The starter mounting provisions shall conform to AND 10262, type XII-S, and shall be suitable for use on an AND20002, type XII-S, engine accessory drive having a clockwise rotation when viewed facing the accessory drive. The overhung moment on the engine pad shall not exceed 625 pound-inches. A quick attach-detach (QAD) device shall be incorporated in the starter.

3.5.7 Ducts. If either the inlet or the exhaust duct is radial, the duct shall be designed to be adjustable in increments of 15° from the vertical. Positioning of the inlet and exhaust shall be independent of each other, if both ducts are radial. The inlet duct shall be located in accordance with FIGURE 2. The exhaust duct location shall be coordinated with the airframe manufacturer and approved by the procuring activity.

3.5.8 Lubrication. The starter shall be capable of running through 300 cycles of starter acceleration and 300 hours of overrunning without requiring additional lubricant. The lubrication system other than specified in 3.5.8.1 shall be in accordance with MIL-STD-838. The lubrication system shall be capable of providing proper lubrication to all starter bearings, seals, and other moving parts at all climb, descent, and cruise altitudes of the aircraft.

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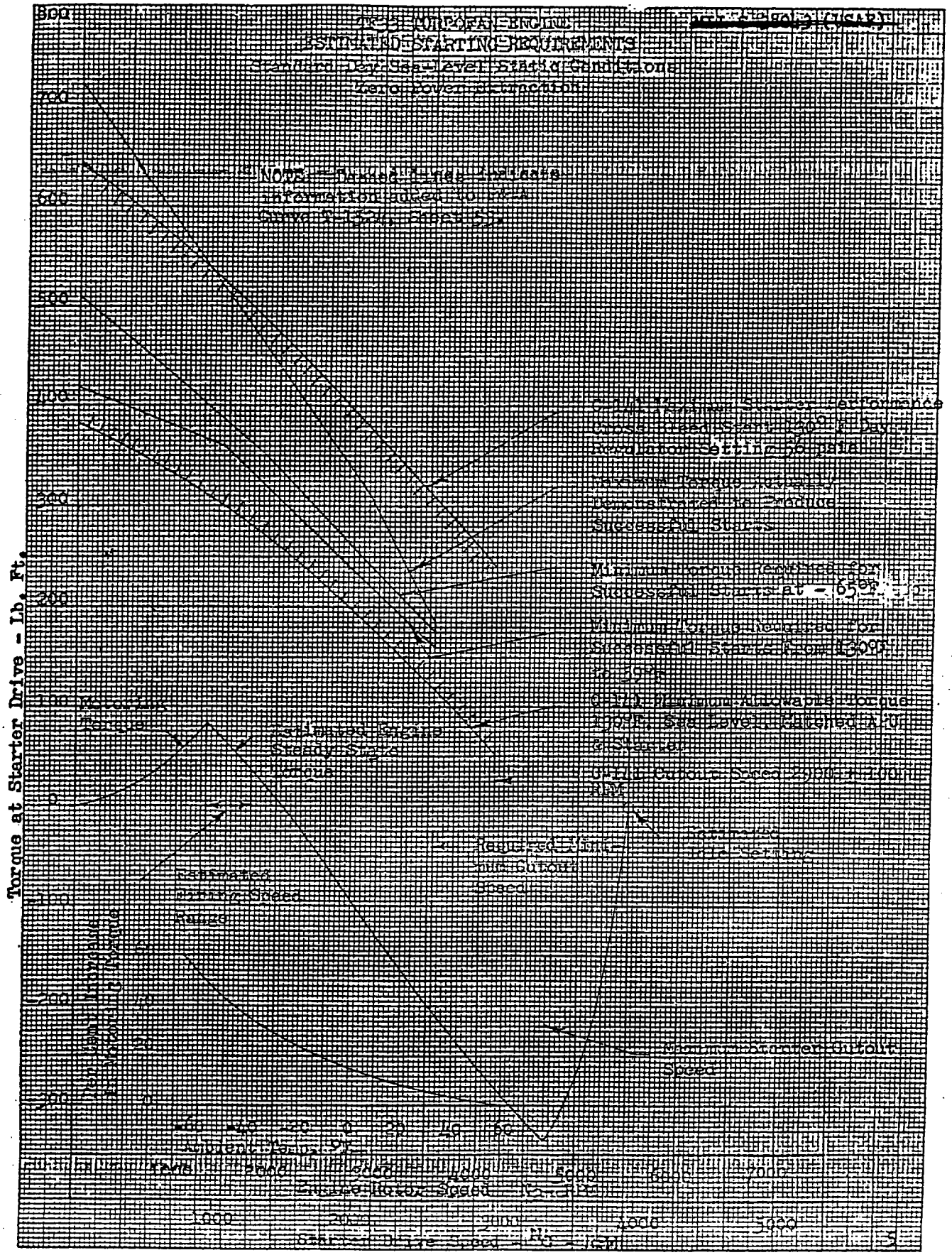


FIGURE 1

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3.5.8.1 Oil Supply. The starter shall have a self-contained lubrication system designed to use oil conforming to MIL-L-7808 and MIL-L-27502. A single lubricant shall be used for the ambient temperature range specified in 3.7.1. The minimum amount of oil necessary for satisfactory starter operation and the oil consumption rate shall be specified in the contractor's model starter specification. Oil conforming to MIL-L-7808 shall be primary lubricant.

3.5.8.2 Oil servicing. All servicing points for lubrication shall be accessible when the starter is completely assembled and installed on the engine mounting pad.

3.5.8.3 Gaging of lubricant. A means of determining the quantity of oil in the starter unit shall be provided in a conveniently accessible location.

3.5.9 Control equipment. Automatic control equipment shall keep the starter within safe operating limits. The automatic control equipment shall be a part of the starter and shall include a control valve which may be removed from the starter for flexibility of installation.

3.5.9.1. Electrical control system. The starter and control valve shall be electrically interconnected so that starter operating throughout the starting cycle will be controlled automatically. The starter shall be capable of being actuated and shut down by a single switch located on the pilot's starting panel. A loss of electrical power shall deactivate the starting cycle. Electrical circuits and parts shall be subject to the approval of the procuring activity.

3.5.9.1.1 Cutout speed. At cutout speed the cutout which on the engine side of the starter shall open the normally closed circuit, from which the control valve is energized, thus closing the valve and shutting off the flow of air to the starter. The electrical circuit shall remain open at all starter output shaft speeds in excess of cutout speed.

3.5.9.1.2 Control valve. The control valve shall have an independent switch which shall close an electrical circuit to light an advisory light on the pilot's starting panel when the valve is in all positions of modulation, and shall open the circuit when the valve is in the fully closed position.

3.5.9.2 Electrical power. Selection of all electrical components provided within the starter and control valve shall be based on the following:

a. A power supply of 18 to 32 volts dc at all ambient temperatures except for a minimum operation down to 14 volts for ambient temperatures up to 185° F.

b. Power consumption shall not exceed 3 amperes inductive at 28 volts.

3.5.9.2.1 Light switch. The advisory light switch in the control valve circuit shall be sized to carry 1-amp current.

3.5.9.3 Control valve. The control valve shown on FIGURE 3 shall be capable of accepting air at the maximum conditions of temperature and pressure specified herein and shall be capable of regulating air at these pressures. The direction of flow through the control valve shall be prominently and permanently marked on the valve. Design of the control valve shall include the following parameters:

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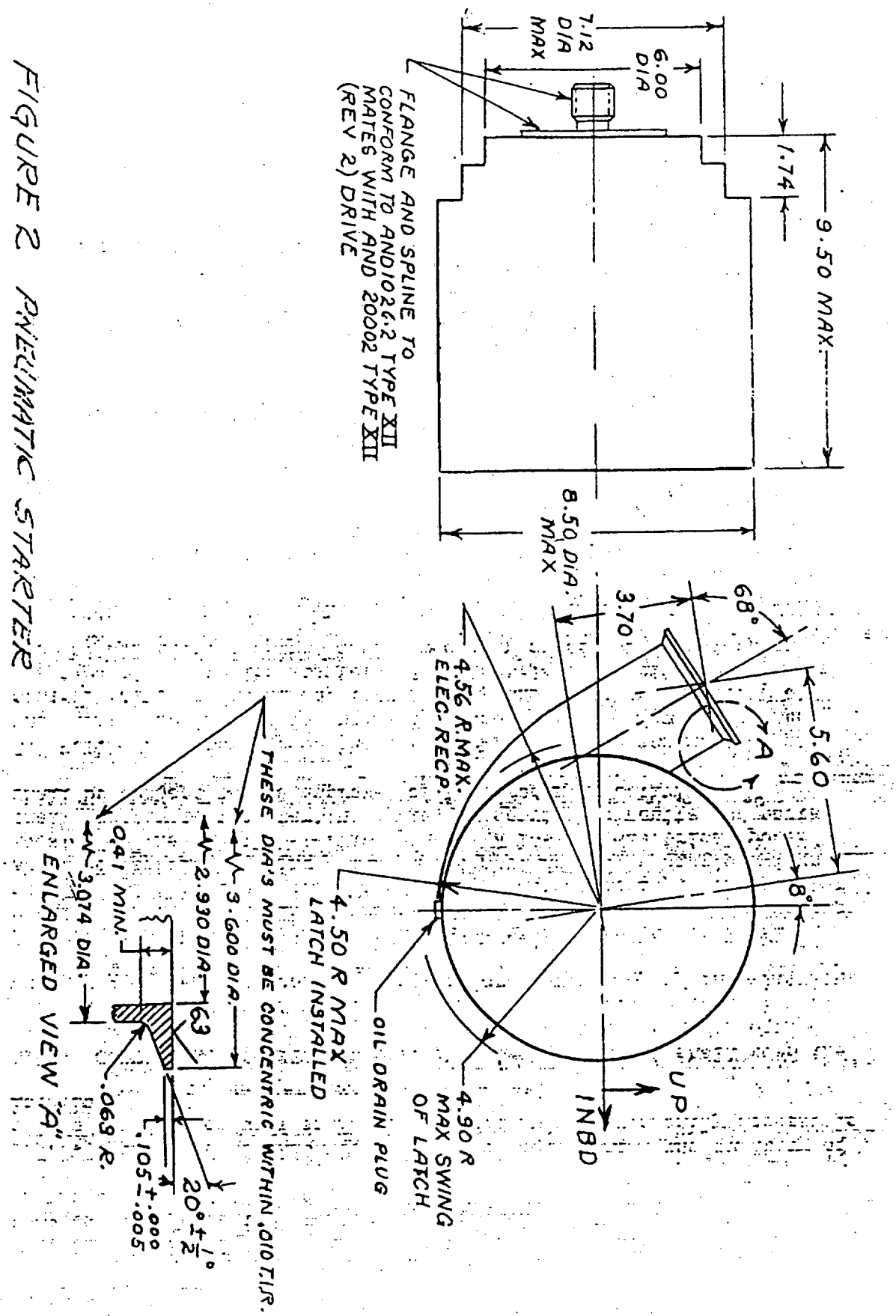
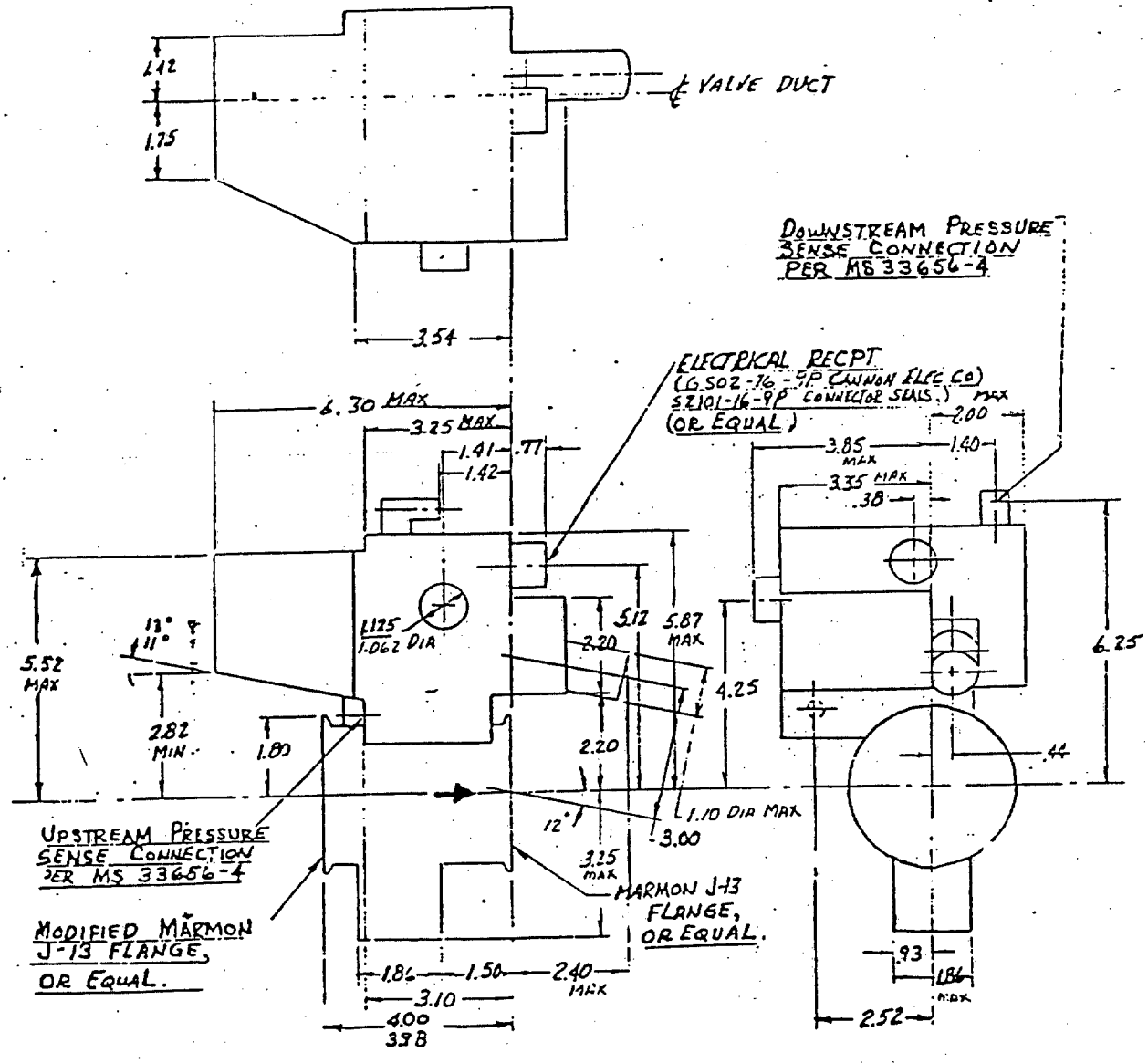


FIGURE 2 PNEUMATIC STARTER

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CONTROL VALVE

FIGURE 3

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a. The control valve shall be a combination air shutoff valve and pressure regulator so designed that it may be remotely located and mounted to reduce the envelope of the starter.

b. The design of the valve and the valve power unit shall be such that the control valve will not be susceptible to clogging of the control orifices by dust, oil, lint or other materials encountered in normal operating conditions. If a filtering device is provided for filtering the control air, the design of the filter shall be as such as to permit proper functioning under the most severe contamination conditions during the normal life of the valve prior to overhaul.

c. The valve shall be capable of maintaining a regulated pressure at any setting within the range of 35.0 to 55.0 psig (50 to 70 psia) discharge pressure. The tolerance on the setting shall not exceed ±5 percent. The initial pressure setting shall be 38 psig. The adjustment shall not be designed for field setting.

d. The control valve shall be so designed that an interruption of electrical power to the valve will cause it to close. An inlet pressure of 16 psig or more shall cause the valve to open when supplied with electrical power. The valve shall also serve as a line shutoff valve. The valve maximum rate of opening shall be such that the rate of pressure increase is limited to 22 psig per second. The control valve opening time shall not impose any excessive shock torques on the starter drive, and the closing rate shall not allow starter rotation faster than the maximum cutout speed. The valve closing time shall be 0.5 second maximum.

e. The valve input shall incorporate a means for indexing the position of the valve to only one position with respect to the duct when the duct has a mating indexing device incorporated.

3.5.9.4 Cutout switch. The starter shall have a cutout speed switch located on the engine side of the overrunning clutch. The turbine side of the starter shall be designed for a fail-safe type of failure in the event of excessive starter turbine wheel overspeed.

3.5.9.5 Electrical requirements. The cutout switch shall be of a hermetically sealed type of sufficient capacity to carry a 5-amp inductive load of 28V dc.

3.5.9.6 Automatic shutoff. The cutout switch shall open a normally closed circuit when the starter drive reaches the designated starter output drive cutout speed as shown on FIGURE 1. When the control valve circuit is opened by the cutout switch, the remote starter control valve shall close and thereby shut off the air supply to the starter.

3.5.9.7 Cutout switch actuating mechanism. The cutout switch actuating mechanism shall be so designed that is impossible to disturb the correct setting of the operation point by normal handling. The mechanism shall not be designed for field adjustment, but may be adjusted at the depot.

3.5.10 Electronic equipment. All electronic equipment shall be in accordance with MIL-E-5400.

3.5.11 Engaging mechanism. An engaging mechanism shall be provided to automatically engage or disengage the starter and the engine under all engine conditions. The engaging mechanism shall permit the started to accelerate from rest and engage the engine at speeds up to the cutout rpm. The cutout speed for the starter shall be 2,900 ±100 rpm.

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3.5.11.1 Free-running turbine. The starter shall be designed to be safe when free running at no load under the maximum pressure and temperature specified at the starter inlet.

3.5.12 Design and fabrication changes. No changes shall be made in the design of parts or materials specified in the approved starter parts list except when such changes are in accordance with the provisions of MIL-STD-480.

3.5.12.1 Material substitutions. Temporary material substitutions may be mad with the approval of the contracting activity.

3.5.12.2 Approval of changes. Approval of changes shall not relieve the contractor of full responsibility for the results of such changes on any of the starter characteristics.

3.5.13 Containment. The starter design shall be such that the rotating parts will not fail when operated at maximum free-run speed. The starter shall be capable of containing fragmentation and maximum-energy type rotor bursts when the unit is operated in accordance with the design criteria specified herein. All high-energy rotors shall be in accordance with CAR 4b, 659. Approval of containment shall be obtained by the starter vendor from the Federal Aviation Agency through the aircraft contractor prior to production unit delivery.

3.5.14 Exhaust screen. The spent gases shall exhaust from the starter turbine without attaching an exhaust from the starter turbine without attaching an exhaust duct. The turbine shall be protected by a heavy-duty metal screen as a containment feature. Means for restraining the screen in position shall be provided to insure that it will not rotate during starter operation.

3.5.15 Rainproofing. The starter and valve shall be designed to operate immediately after or during a severe rainstorm, assuming that the cowl access panels have been removed and that the components are exposed to the elements.

3.5.16 Safetying. All internal and external threaded parts shall be positively locked or safetied by wiring or self-locking means. Safety wire shall be applied in accordance with MS33540 and shall conform to MS20995. Self-locking nuts shall be in accordance with MIL-N-25027. Star washers or jam nuts shall not be used as a sole means of safetying. The application of lock washers shall be governed by the limitations of AND10476.

3.5.17 Reliability. The starter and control valve shall be of such design as to achieve the highest operational commensurate with design requirements.

3.5.17.1 Reliability data. The manufacturer shall in include in his proposal an estimate of the resent unit failures, based on 100 units that can be expected during each 50-hour interval from 0 to 1,000 hours in normal fleet operation. This may be substantiated by service records of the unit or a similar unit in equivalent or similar usage. If available, service records of starter units that have containment features and operate in relatively high ambients should be utilized to substantiate reliability.

3.5.17.2 Reliability test. Reliability test may be required dependent upon the results of preproduction and other tests.

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3.6 Construction. The starter shall be so constructed that no parts will work loose in service. It shall be built to withstand accelerations up to 15g resulting from the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Maintenance. The starter shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases, with the exception of the initial control valve setting as denoted in 3.5.9.3c. It shall not be possible to disturb the setting of the speed switch or other critical components by mere handling of these parts. Consideration shall be given to the routine maintenance items, such as checking the oil level at frequent intervals and checking the screens in the control system of the regulator valve for clogging.

3.7 Performance. The starter shall satisfy the performance requirements specified in section 4 when subjected to the following tests:

a. Calibration Tests

b. Endurance Tests

c. Environmental Tests

3.7.1 Pressure and temperature. The starter shall operate satisfactorily through an ambient temperature range of -65° to +400° F continuous and a maximum of +500° F for a period not exceeding 30 minutes. The starter shall operate satisfactorily when supplied with air at maximum temperature and pressure conditions.

3.7.2 Torque. The starter shall not exceed the maximum torque specified on FIGURE 1 when supplied with air from an operating gas turbine engine which delivers 245 psia and 900° F, and unlimited airflow to the starter control valve inlet with the valve set at the maximum regulated pressure of 41 psig.

3.7.3 Auxiliary power unit conditions. The starter shall provide the minimum torque specified on FIGURE 1 when supplied with air from an auxiliary power unit. The output parameters (mass flow, pressure, and temperature) shall be specified by the airframe manufacturer and approved by the procuring activity.

3.7.4 Age cart conditions. The starter shall provide the minimum torque specified on FIGURE 1 when supplied with air from an AGE Cart. The output parameters (mass flow, pressure, and temperature) shall be specified by the airframe manufacturer and approved by the procuring activity.

3.7.5 Cold weather operations. Due to the thermal lag in the pneumatic system piping, it is possible for the air supplied to the starter to approach -65° F. The starter shall be suitable for operation at the maximum regulated pressure and a minimum temperature of -65° F at the starter inlet.

3.7.6 Maximum bleed duct conditions. The maximum bleed duct conditions shall be as follows:

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Ambient	-65° F
Bleed air temperature	+550° F
Bleed air pressure	260 psia
Ambient	+130° F
Bleed air temperature	+900° F
Bleed air pressure	245 psia

3.7.7 Design pressures. The design pressures shall be as follows:

Valve operation	260 psia maximum
Valve closed	260 psia design 390 psia proof 650 burst

3.7.8 Inlet temperature. The inlet temperature shall be 900° F maximum.

3.7.9 Valve solenoid. The actuating solenoid from the valve power unit shall be of the continuous-duty type and shall be capable of operating in a 400° F ambient for a period of 24 hours (no flow-thru compressor bleed ducting or starter piping).

3.7.10 Altitude of operation. The starter shall be capable of performing satisfactory starts at altitudes from sea level to 10,000 feet, and shall be capable of overrunning at altitudes from sea level to 50,000 feet.

3.8 Duty cycle. The duty cycle shall consist of a, b, and c when operating at the following conditions: (1) 400° F ambient, (2) 3-hour soak at 400° F with starter inlet air conditions at 900° F, and (3) maximum regulated pressure.

a. 1 minute on, 30 seconds off, 1 minute on, 30 seconds off, 1 minute on, 30 seconds off.

b. 1 minute on, 5 minutes off.

c. 1-1/2 minutes on, 10 minutes off.

3.9 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

3.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.11 Weight. The total weight in pounds of the starter and the valve, including starter lubricants, shall be specified in the contractor's model specification.

3.12 FINISHES AND PRODUCTIVE COATING

3.12.1 Anodizing. All aluminum alloy parts shall be anodized in accordance with MIL-S-8625.

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3.12.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.12.3 Painting. All exterior exposed surfaces except corrosion resistant steel, working surfaces, and threads shall be finished with one coat of primer conforming to TT-P-1757 followed by two coats of black enamel conforming to TT-E-489, type III.

3.13 Identification of products. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13.1 Nameplate. A nameplate shall be permanently attached to the starter and control valve and shall conform to MIL-P19834. In addition to the information required by MIL-STD-130, the following shall be included:

a. Starter Nameplate

Starter, Pneumatic, Aircraft Engine, Air Turbine STU-16/A

b. Valve Nameplate

Maximum Inlet Air Temperature 900° F
Maximum Allowable Inlet Air Pressure 260 psia
Regulated Pressure Delivery 38 ±3 psig

3.13.2 Rotation. The direction of the starter output shaft shall be marked in a conspicuous position.

3.13.3 Synthetic rubber parts. Equipment and assemblies containing synthetic rubber parts shall be marked in accordance with MIL-STD-1523.

3.14 Workmanship. The starter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.14.1 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.14.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.14.3 Cleaning. The starter shall be thoroughly cleaned and loose spattered, or excessive solder, metal chips, and other foreign material are removed after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction tests (see 4.4)

b . Acceptance tests (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specification.

4.3.1 MEASUREMENTS

4.3.1.1 Temperature. All temperatures shall be measured in degrees Fahrenheit.

4.3.1.2 Speed. Speed shall be measured by a tachometer, accurate within .50 of 1 percent in the cutout speed and free-running speed range.

4.3.1.3 Air. The amount of air consumed shall be measured in pounds per minute.

4.3.2 APPARATUS

4.3.2.1 Instrument calibration. All test instruments and equipment upon which the accuracy of the test results depend shall be calibrated frequently enough to assure laboratory accuracy.

4.3.2.2 Flywheel. The flywheel used when testing the starter shall have a sufficient polar moment of inertia to result in an acceleration time of not less than 20 seconds from rest to starter cutout speed when the starter is delivering the guaranteed minimum torque.

4.3.3 Inspections and adjustments. Unless otherwise specified by the contractor and approved by the procuring activity, all services, such as lubrication and general external visual inspection, shall be made

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after every 300 cycles. Disassembly of any part of the starter to vicing, shall be at the option of the procuring activity. At convenient times prior to or during the progress of the test and during the teardown inspections, the starter shall be examined to determine if it conforms to specification requirements.

4.3.4 Part failures. If, during the preproduction testing of a starter a part fails, this part shall be replaced with a part of improved design or a part of more suitable material, unless the procuring activity authorizes the installation of a new part of the original design and material for one which failed due to faulty material and workmanship.

4.3.5 Tolerances. Unless otherwise specified, the tolerance on temperatures shall be +25° F, the tolerance on pressures shall be ±5 psi, and the tolerance on speed shall be ±2 percent (rpm).

4.4 PREPRODUCTION TESTING

4.4.1 Test samples. The test samples shall consist of two starters and two control valves representative of the production equipment. They shall be tested at the contractor's plant under the supervision of the procuring activity.

4.4.1.1 Detail design data. The following data shall be submitted to the procuring activity:

a. Outline drawings showing pad and drive upon which the starter will mount and the size and shape of the starter together with the clearance volume required for mounting the starter and the control valve.

b. Assembly drawing.

c. Drawings showing size, location, and flange details of the starter inlet and outlet ducts.

d. Drawings showing size, location, and flange details of control valve inlet and outlet ducts.

e. Air pressures and temperatures at the rated and maximum conditions for the starter.

f. Weight of the entire starter.

g. The guaranteed minimum torque delivered by the starter when supplied with air at rated conditions.

h. The air consumption of the starter when operation at rated condition.

i. The free running speed of the starter when supplied with air at the rated condition.

j. Electrical power required by the starter.

k. The specification of any lubricant required for servicing the starter.

l. The size of the flywheel that can be accelerated to cutout speed in the acceleration time (±2 seconds) when the starter is supplied with air at rated conditions.

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m. The minimum load at which the replaceable shear section in the starter drive will fail.

4.4.1.2 Preproduction tests. The manufacturer shall furnish the following performance curves along with or prior to submittal of a sample starter for preproduction tests.

a. A chart presenting curves of guaranteed minimum torque versus speed from standstill to cutout speed.

b. A chart presenting curves of guaranteed maximum air consumption versus speed from standstill to cutout speed.

c. A chart on starter control performance presenting curves of all variables versus speed.

d. A chart presenting curves of starter free run characteristics at various pressure ratios and inlet air temperatures.

4.4.1.2.1 Back pressure. All charts shall present curves at maximum temperature, rated temperature, 80 percent rated absolute temperature, and 60 percent rated absolute temperature for air at rated pressure and a back pressure of 15 psia.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6. The tests shall be conducted as follows:

a. Starter Sample 1 and Control Valve Sample 1

(1) Examination of product

(2) Initial run

(3) Calibration tests

(4) Endurance tests

(5) Final calibration

(6) Teardown inspection

(7) Containment test

b. Starter Sample 2 and Control Valve Sample 2

(1) Examination of product

(2) Initial run

(3) Calibration tests

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(4) Environmental tests

(5) Final calibration

(6) Teardown inspection

4.5 Acceptance tests. Acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each starter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Initial run

4.6 TEST METHODS

4.6.1 Examination of product. The starter shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and markings.

4.6.2 INITIAL RUN

4.6.2.1 Control valve. The control valve shall be subjected to the following to determine that specified performance parameters have been met.

a. Three cycles of opening and closing (minimum open time 1 minute) with the inlet air at rated conditions of pressure, temperature, and air flow (245 psia, 900° F). The pressure drop across the valve shall not exceed the value determined in 4.6.3.2.

b. Three cycles of opening and closing (minimum open time 1 minute) with inlet air at maximum conditions. The outlet pressure shall not exceed the allowable inlet conditions of the starter.

4.6.2.2 Starter and control valve. The complete starter and control valve shall demonstrate guaranteed performance when subjected to the following:

a. Starter airflow shall be checked with the starter in a stalled condition using air at rated conditions.

b. Three cycles of acceleration the flywheel of 4.3.2.2 from standstill to cutout speed using air at rated conditions. Performance shall be calculated by recording speed versus time at 3-second intervals, using flywheel moment of inertia and making allowance for friction losses on at least one of these cycles.

c. The starter output shaft shall be driven at maximum engine speed (or a minimum of 2-1/2 times minimum cutout speed) for a period of 15 minutes to demonstrate satisfactory operation of the overrunning clutch.

4.6.2.3 Inspection after initial run. It shall not be necessary to disassemble starters for inspection after the initial run, provided means of detecting malfunctioning starters are devised and used during the tests.

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4.6.2.4 Final run. The final run shall consist of repeating the test specified in 4.6.2.1a or 4.6.2.1b.

4.6.3 CALIBRATION TESTS

4.6.3.1 Starter calibration. Torque at the engine drive and starter air consumption shall be determined at 500-rpm increments from zero to 2,500 rpm, and then at 200-rpm increments up to maximum cutout speed using air input to the control valve at the following conditions:

a. 245 psia to 900° F

b. 260 psia at 555° F

c. 205 psia at 750°F

d. To be specified b y airframe manufacturer and approved by procuring activity (see 3.7.3) for the -65° F ambient condition.

e. To be specified by airframe manufacturer and approved by procuring activity (see 3.7.3) for the conditions of 60° F ambient.

f. To be specified by airframe manufacturer and approved by procuring activity (see 3.7.3) for the conditions of 130° F ambient.

4.6.3.1.1 Air Consumption. Torque and air consumption versus speed curves shall be obtained for each of the conditions specified in 4.6.3.1. The torque shall equal of exceed the value specified in the performance curves of 4.4.1.2.

4.6.3.2 Control calibration. Calibration tests shall determine the relation between control component input variables and control component outputs. Only those components for which the input-output relations are a part of the function of the component shall be subject to calibration. The calibration tests shall be devised by the contractor and approved by the procuring activity.

4.6.3.3 Starter speed switch calibration. The sped switch shall be calibrated to determine that the cutout speed is between 2,800 and 3,000 starter rpm.

4.6.4 Starter drive shear test. Five shear sections, identical to design to those utilized in the starter, shall be statically loaded to failure. Failure shall occur at a torque value not exceeding 1,050 ±50 pound-feet.

4.6.5 Transient torque. Transient torque peaks shall be determined in accordance with SAE report AIR 62. The backlash and stiffness of the engine tear train which are 1° 23.2' and 192,500 pound-inch/radian, respectively, shall be simulated on the impact torque test rig. The torque shall be determined with the control valve installed and with inlet conditions of 245 psia and 900° F. The airflow shall be sufficient to maintain a choked nozzle.

4.6.6 ENDURANCE TESTS

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4.6.6.1 Acceleration. The starter and the valve shall be subjected to a 1,000-cycle test at sea level pressure altitude and at the ambient temperature range indicated in TABLE 1. Each cycle shall con consist of engagement, acceleration of the flywheel specified in 4.3.2.2 from rest to cutout rpm, and disengagement. A rest period to enable the starter oil temperature to return to approximately 20° F above ambient temperature may be taken between cycles. At the end of each acceleration cycle, the output shaft shall be driven above the cutout speed of the starter, or sufficient data shall be obtained to demonstrate complete disengagement of the starter from the loading mechanism. Following each 25 acceleration cycles, the starter output shaft shall be externally driven at 7,359 rpm for a period of 75 hours for an accumulated total of 3,00 hours. During this overrunning, the starter ambient and starter pad temperature shall be maintained at 350° to 400° F. During each 75-hours period when the starter output shaft is externally driven, the starter shall be inclined to simulate a 12° aircraft nose-up altitude for 5 hours; at 14°aircraft nose-down altitude for 5 hours; and a 2° nose-up altitude for 65 hours. During this test, it shall not be necessary to add oil to the starter in less than 300-hour intervals (see 3.5.8).

4.6.6.2 Free run. With no load on the starter drive, the starter shall e accelerated 10 times from test to maximum speed using air at rated conditions. The starter shall be subjected to a 1 minute no load free run test at regulated pressure followed by one simulated start. Following the 1 minute free run test, the starter shall be free run to failure.

Period	No of Cycles	Ambient Temp (°F)	Ambient Temp (°F)	Inlet Press (Psia)
	150	60 - 90	Note 1	Note 1
I	75	60 - 90	900 ± 25	245
	75	60 - 90	800 ± 25	260
	100	400	Note 1	Note 1
II Note 2	50	400	900 ± 25	245
	50	400	800 ± 25	260
	250	60 - 90	Note 1	Note 1
III	125	60 - 90	900 ± 25	245
	125	60 -90	800 ± 25	260

Note 1: This condition to be specified by the airframe manufacturer and approved by the procuring activity.

Note 2. During this test period, the capability of the starter to operate for periods of 30 minutes a 500° F ambient shall be demonstrated.

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4.6.6.3 Engaging tests. The flywheel shall be driven at engaging speed and the starter engaged and accelerated to cutout speed. This cycle shall be for engaging speeds from 500-rpm throughout the cutout speed in 500-rpm increments.

4.6.6.4 Starter no-load cutout. The starter shall be accelerated 10 times at no load with maximum bleed conditions, 245 psia and 900° F, at the starter bled air inlet. This test shall demonstrate the ability of the cutout switch to close the valve before the starter reaches excessive speed. The starter shall sustain no damage during this test.

4.6.6.5 Starter duty cycle. The duty cycles defined in 3.8a shall be repeated 5 times continuously, with no additional time between cycles, for a total of 15 starts. Each of the duty cycles defined in 3.8b and 3.8c shall be repeated for 15 times continuously with no additional time between cycles. Ambient air temperature during this test shall be 375° /f minimum. This test shall be conducted with the start loaded by a dynamometer. The load shall be such that the starter shaft speed will stabilize at 1,200 to 1,500 rpm when then starter inlet air is 900° F and at maximum regulated pressure. One duty cycle shall consist of bringing the dynamometer up to stabilized speed within 10 seconds and maintaining this stabilized speed for the remainder of the ON time.

4.6.6.6 Containment test. A starter shall be subjected to an inlet air temperature of 900° F ±25° F. At this temperature, an induced turbine wheel burst shall be effected at a speed equal to or in excess of the maximum free run speed when utilizing inlet air in accordance with 3.7.6. This test shall be conducted only after completion of teardown inspection specified in 4.6.9.

4.6.6.7 VALVE TESTS

4.6.6.7.1 Pressure test. The valve shall be pressure tested at the following inlet pressures and temperatures for a period of 30 minutes each:

a. 245 ±5 psia at 900°±10° F

b. 260 ±5 psia at 555°±10° F

The valve shall be operated and shall open 30 times each at conditions a and b. No permanent damage or distortion of any part shall result from this test when the valve is either open or closed.

4.6.6.7.2 Valve operation. Air under rated conditions shall be supplied to the valve, and the valve used to regulate this air to the starter. After the valve position has stabilized, the air and the electric power supplied to the valve shall be cut off in that order. The position indicator on the valve shall show that the valve has closed. With the valve closed, air at 16 psig and 250° shall be supplied to the valve when the valve is closed. The starter switch shall then be turned on the valve shall open.

4.6.6.7.3 Valve leakage. With the valve in the closed position, the upstream side shall be subjected to a pressure of 245 psia and 900° F. The leakage past the valve and through the control orifices shall not exceed 0.25 ppm.

4.6.6.7.4 Airflow test. The combination pressure regulator and shutoff valve shall be mounted on a line with an orifice simulating the starter and subjected to airflow tests with pressures and temperatures as specified in 4.6.3.1. The valve shall demonstrate that it will open under these pressures and regulate within

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5 percent of the regulated pressure settings. With the control valve closed, the upstream side shall be pressurized to 30 psia and held at 30 psia maximum to demonstrate that the valve will function normally (open and closed) down to 30 psia minimum pressure. The pressure drop through the valve shall be determined at the pressures and temperatures specified in 4.6.3.1.

4.6.7 ENVIRONMENTAL TESTS

4.6.7.1 Cold operation test. The starter shall be subjected to 10 cycles of operation as follows:

a. Soak the starter at -60° to -70° F for 20 hours except that after the first cycle, the procuring activity may permit reduction of the soak period to not less than 5 hours.

b. Engage the starter to the flywheeler a dynamometer to load of at least equivalent moment of inertia, accelerate the flywheel or dynamometer to not less that 50 percent of disengagement speed, and continue to operate the starter for 90 seconds at maximum torque corresponding to the stabilized speed which shall be not less than 50 percent nor more than 70 percent of disengagement speed. This test shall be accomplished 10 minutes after removing the starter from the low temperature area if it is not possible to conduct all parts of the test in that area. The air supply shall be unheated shop air at 90 psi minimum.

c. Soak the starter for 15 minutes at -60° to -70° F and repeat procedure b.

d. Repeat procedure c.

4.6.7.2 Humidity test. The humidity test shall be conducted in accordance with Procedure I - Natural of MIL-STD-810. The test period shall be 15 cycles. The unit shall be so mounted as to simulate installation in the aircraft. The test chamber shall be vented to the atmosphere to prevent the build-up of pressure. Prior to the starting of the test period, the chamber shall be between 20° and 38° C (68° and 100° F) with uncontrolled humidity. During the first 2 hour period, the temperature shall be gradually raised to 71° C (160 °). The temperature shall be maintained for the next 6 hour periods. The velocity of the air throughout the test area shall not exceed 150 feet per minute. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 20° to 38° C (68° to 100° F) which shall constitute one cycle. The relative humidity throughout the cycle shall be 95 percent. Steam or distilled water having PH value between 6.5 and 7.5 sy 25° C (77°) shall be used to obtain the desired humidity. After completion of the 15th cycle, the unit shall be subjected to an ambient temperature of 65° ±5° F for 5 hours without drying, shaking, or otherwise removing condensed water from the unit. Immediately after the cold soak period, the starter spline drive shall be externally driven for 2 minutes at maximum engine speed (or a minimum of 2.50 times the minimum cutout speed). The drive shall overrun the starter turbine in accordance with the design of the starter. Immediately thereafter, the starter shall be used to accelerate the flywheel from rest to cutout speed. The starter shall be examined and tested or operated as necessary to determine that no harmful effects have resulted from this test.

4.6.7.3.1 Starter and control valve (operating). During the acceleration test, 4.6.6.1, period II, the second group of 50 cycles (900° F inlet temperature, 245 psia inlet pressure) the equipment shall be placed in a test chamber equal to MIL-C-9436 and the sand and dust density raised and maintained at 0.1 to 0.25 grams per cubic foot, the temperature shall be 25° (77°F), the air velocity shall be 2,500 ± 500 feet per minute. One hundred percent of the sand and dust shall pass through a 100 mesh screen, U.S. Standard Sieve Series. The sand and dust for use in the procedure is commercially known as "140 mesh silica

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flour". Sand and dust (140 mesh silica flour) produced by the Fenton Foundry Company, Dayton OH and Ottwa Silica Company, Ottana, IL or equal is satisfactory for use in the performance of these tests. The test chamber shall be vented to the atmosphere. The relative humidity shall not exceed 30 percent at any time during the tests.

4.6.7.3.2 Starter and control valve (non operating). In addition to the above test, a sand and dust test shall be performed, while the starter and control valve are inoperative. The test procedure shall be as specified in 4.6.7.3.1 with a duration of 3 hours with sand and dust density at 0.1 to 0.5 grams per cubic foot within the test space. One hundred percent of the sand and dust shall pass through a 100 mesh screen, U.S. Standard Sieve Series.

4.6.7.4 Fungus-resistance test. A fungus test shall be conducted in accordance with Method 508.3 of MIL-STD-810 on a representative starter and a representative control valve. If no fungus nutrients exist in either assembly, a notarized statement to this effect will be accepted in lieu of testing.

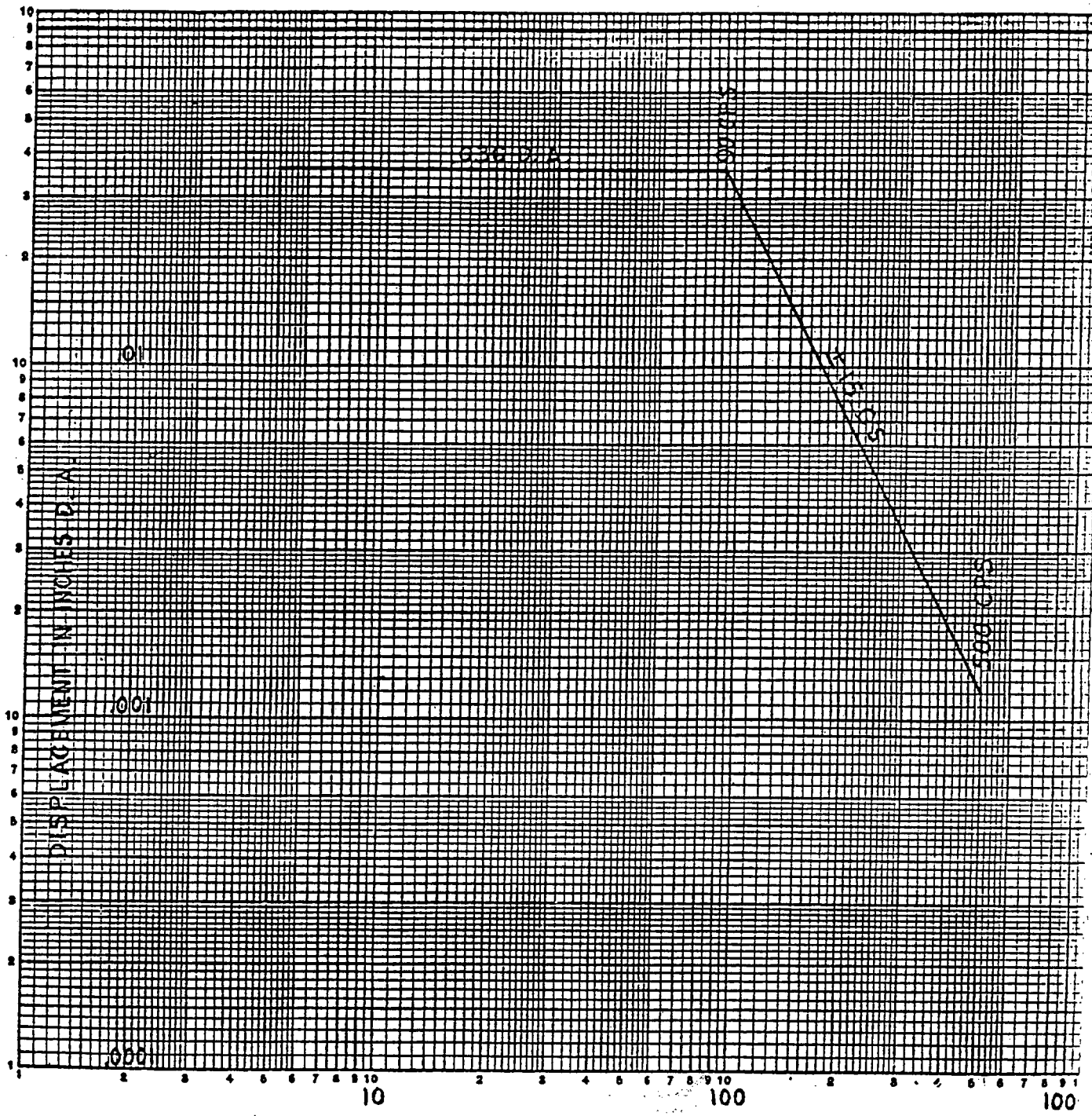
4.6.7.5 Rain test. The starter and valve shall be mounted in the test chamber simulating the aircraft installation and subjected to a rain test. The rain test chamber temperature shall be uncontrolled except as regulated by water introduced as rain, throughout the test period. A simulated rainfall 4 ± 1 inch per hour as measured at the surface of the equipment by a U.S. Weather Bureau type gauge shall be produced by means of a water spray nozzle of such design that the water is emitted in the form of droplets having a minimum diameter of 1. millimeters. The temperature of the water shall be uncontrolled if the water supply temperature is between 11° to 20° C (51.8° to 68° F). The direction of rainfall shall be capable of variation up to 45° from the verticle. The rainfall shall be dispensed uniformly over the test area within the limits as specified above. At the completion of the test, the control valve shall be energized and the starter accelerated to cutout speed in not less than 20 seconds using air at a temperature of 50° to 120° F. This test shall be run at no load. After the operational test, the equipment shall be examined for evidence of water penetration or deterioration.

4.6.7.6 Salt fog test. The starter and valve shall be subjected to a salt fog test in accordance with MIL-STD-810. The length of the test shall not be less than 168 hours. At the completion of the test, the control valve shall be energized and the starter accelerated to cutout sped in not less than 20 seconds using air at a temperature of 50° to 120° F. This test shall be run at no load. The equipment shall be inspected for corrosion damage prior to operation. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water. The equipment operation and inspection shall be performed with 1 hour after termination of the salt fog test. Equipment shall again be operated and inspected 48 hours later.

4.6.7.7 High temperature test. The portion of the valve which contains the hot airflow shall be soaked at 900° F for 6 hours with the remainder of the valve surrounded by air at 400° F. At the end of the soak period, the valve shall be energized in accordance with the duty cycle specified in 3.8a. The high temperature soak and valve energizing cycle shall be repeated for a total of five times. This test shall demonstrate that the valve is not adversely affected by heat transferred from the hot engine compressor blood piping during normal flight operations.

4.6.7.8 Vibration test. The starter and valve shall be subjected to a vibration test utilizing FIGURE 4 of this specification. The specimens shall be attached to a rigid fixture capable of transmitting the vibration conditions specified herein. Attachment of the specimen to the fixture shall be made through the service mounting which represents dynamically the most adverse of alternate service mountings possible. Each

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CURVE I

FIGURE 4 VIBRATION TEST REQUIREMENT

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resonant and cycle period shall be divided into three equal parts, the first part being conducted at -54°C *-65°F), the second part at room temperature, and the third part at 71° (160°F). Tests shall be conducted under both the resonant and cycling conditions specified. The order in which the resonance and cycling tests are performed shall be optional.

Vibration Test Schedule

Number of Resonsances	0	1	2	3	4
Total Vibration Time at Resonance*	-	30 Min	1 Hr	1 1/2 Hr	2 Hr
Cycling Time	3 Hr	2 1/2 Hr	2 Hr	1 1/2 Hr	1 Hr

* 30 minutes at each resonance

Resonance modes of the test specimen shall be determined by varying the frequency of applied vibration slowly through the specified range at vibratory accelerations not exceeding those in FIGURE 4. Individual resonance surveys shall be conducted with vibration applied along each axis of any set of three mutually perpendicular axis of the test specimen. The test specimen shall be vibrated at the indicated resonant conditions for the periods shown in the vibration test schedule and with the applied amplitudes of vibratory accelerations in FIGURE 4. The vibrator acceleration applied to specimens weighting more than 50 pounds may be decreased by ±1g for each pound increment of weight more than 50 pounds to a value no lower than + 5g. These periods of vibration shall be accomplished with vibration applied along each of three mutually perpendicular axis of vibration. When more than one resonance shall be sustained for the period shown in the applicable portion of the vibration test schedule. If more than four resonances are encountered with vibration applied along any one axis, the four most severe resonances shall e chosen for test. There shall be no resonant peaks within the range 25 to 170 cps. If they are unavoidable, the amplification factor shall not exceed 1.25 during the vibration test. Individual parts of the starter and valve shall be considered acceptable if no mechanical failure occurs and the unit operates satisfactorily after the vibration test.

4.6.8 Final calibration. Torque at the engine drive shall be determined at 500-rpm increments, from zero to 2,500 rpm, and then 200-rpm increments up to maximum cutout sped, utilizing the pressure and temperature conditions specified under 4.6.2.1. Pressure limits shall be ±5 psia. Temperature limits shall be ±25° F. The performance of the starter shall not have decreased more than 5 percent from the guaranteed minimum torque valves specified in 4.4.1.2.

4.6.9 Teardown inspection. After completion of the endurance and environmental tests, the starter shall be completely disassembled for examination of all parts and for measurements as necessary to disclose excessively worn, distorted, or weakened parts. These measurements shall be compared with the contractor's drawings or with similar measurements made prior to the preproduction test, if available. The preproduction tests shall be considered complete when every major part in the starter has been subjected to and has completed the preproduction tests without damage.

4.7 Retest. Rejected starters shall not be submitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

5. PACKAGING

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5.1 Preservation-packaging. Preservation-packaging shall be level A, C, or Industrial as specified. (see 6.2)

5.1.1 Level A. Unless otherwise specified by the contracting activity, each starter shall be individually preserved and packaged according to MIL-STD-2073-1. The method of preservation shall conform to submethod II-A of MIL-P-116.

5.1.2 Level C. Each starter shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipments from supply source to the first receiving activity for immediate use.

5.1.3 Industrial. The industrial preservation of starter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level(s) A, B, C. The level of packing shall be specified by the contracting activity. The level of packing shall be accomplished in accordance with the requirements outlined in MIL-STD-2073-1.

5.2.2 Industrial. The packaged starter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of the specifications.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The STU-16/A starter covered by this specification is intended for use in starting aircraft gas turbine engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Levels of packaging and packing desired

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Subject term (key word) listing.

Air turbine starter
Engine starter
Pneumatis starter
STU-16/A

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing Activity:
AIR FORCE -71

Project Number
2995-F176

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-38043	

3. DOCUMENT TITLE
Starter, Engine, Air Turbine STU-16/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
OC-ALC/TICLA	(1) Commercial (405) 736-5960 (2) AUTOVON 336-5960
c. ADDRESS (Include Zip Code) 3001 Staff Dr Tinker AFB, OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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